
🔍 Architecture Explanation

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## 🔍 Architecture Explanation

The system follows a **modular, scalable automation architecture** designed for location-based business data collection and engagement.

###1️⃣User / Admin Input
- User provides:
  - Business category (e.g., real estate, clinics, shops)
  - Target geographic location
- This input acts as the trigger for the automation pipeline.

###2️⃣n8n Workflow Engine
- n8n orchestrates the complete automation flow.
- Handles:
  - Workflow triggers
  - API calls
  - Conditional logic
  - Error handling
- Ensures the system remains **low-code, reusable, and scalable**.

###3️⃣Serper API (Google Maps Data Source)
- Fetches business listings from Google Maps.
- Returns structured results such as:
  - Name, address, phone number, website, ratings, and hours.
- Supports pagination for large datasets.

###4️⃣Data Parsing & JSON Processing
- Raw API data is:
  - Cleaned
  - Normalized
  - Converted into structured JSON objects
- Duplicate and invalid entries are filtered.
- Unique identifiers are assigned for data integrity.

###5️⃣Google Sheets (Central Database)
- Acts as the primary data store.
- Automatically appends new business records.
- Enables:
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  - Easy filtering and analysis
  - Dashboard creation
  - CRM-style lead tracking

6️⃣ Engagement Layer
- Connected landing page enables:
  - Increased engagement
  - Lead capture
- Automation supports:
  - Auto-replies
  - Faster customer response
  - Improved lead conversion

7️⃣ Business Analytics & Lead Management
- Data from Google Sheets is used for:
  - Market research
  - Outreach campaigns
  - Lead scoring and prioritization
- Can be extended to BI dashboards or CRM tools.